

For a reaction of order n , the integrated form of the rate equation is $(n-1) \cdot K \cdot t = (C_0)^{1-n} - (C)^{1-n}$, where C_0 and C are the values of the reactant concentration at the start and after time ' t '. What is the relationship between $t_{3/4}$ and $t_{1/2}$, where $t_{3/4}$ is the time required for C to become $C_0/4$.

✓ (A) $t_{3/4} = t_{1/2} \cdot [2^{n-1} + 1]$

(B) $t_{3/4} = t_{1/2} \cdot [2^{n-1} - 1]$

(C) $t_{3/4} = t_{1/2} \cdot [2^{n+1} - 1]$

(D) $t_{3/4} = t_{1/2} \cdot [2^{n+1} + 1]$

$$(n-1) K t = \frac{1}{(C_0)^{n-1}} - \frac{1}{(C)^{n-1}}$$

$$t_{3/4} = \frac{1}{(n-1) K} \left[\frac{1}{(C_0)^{n-1}} - \frac{1}{\left(\frac{C_0}{4}\right)^{n-1}} \right]$$

$$= \frac{1}{(n-1) K} \left[\frac{1 - 4^{n-1}}{C_0^{n-1}} \right]$$

$$t_{1/2} = \frac{1}{(n-1) K} \left[\frac{1}{(C_0)^{n-1}} - \frac{1}{\left(\frac{C_0}{2}\right)^{n-1}} \right]$$

$$t_{1/2} = \frac{1}{(n-1) K} \left[\frac{1 - 2^{n-1}}{(C_0)^{n-1}} \right]$$

$$\frac{t_{3/4}}{t_{1/2}} = \frac{1 - 4^{n-1}}{1 - 2^{n-1}}$$

$$= \frac{(1)^2 - (2^{n-1})^2}{1 - 2^{n-1}}$$

$$= \frac{(1 + 2^{n-1})(1 - 2^{n-1})}{1 - 2^{n-1}}$$

$$= (1 + 2^{n-1})$$

In a certain first-order reaction, B^{n+} is getting converted to $B^{(n+4)+}$ in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing agent which reacts only with B^{n+} and $B^{(n+4)+}$. In the process, it converts B^{n+} to $B^{(n-2)+}$ and $B^{(n+4)+}$ to $B^{(n-1)+}$. At $t = 0$, the volume of reagent consumed is 25 ml and at $t = 10$ minute, the volume used is 32.5 ml. The rate constant for the conversion of B^{n+} of $B^{(n+4)+}$ is ($\ln 2 = 0.7$, $\ln 5 = 1.6$)

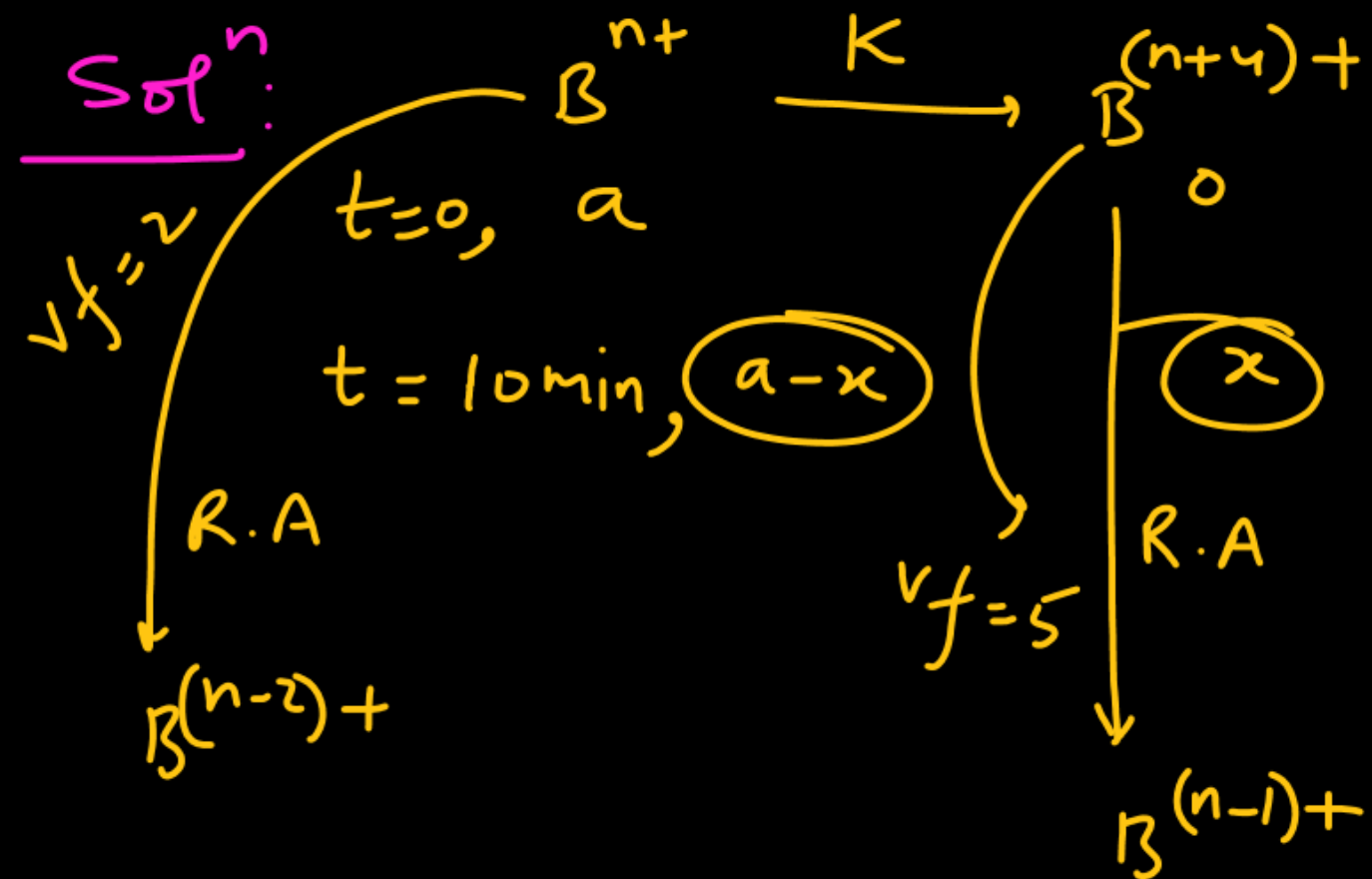
(A) 0.2 min^{-1}

(B) 0.8 min^{-1}

(C) 0.02 min^{-1}

(D) 0.08 min^{-1}

Imp



[1st order]

At $t = 0$, Mo. of g. eq. of $B^{n+} = a \times 2 \propto 25$

$a \propto 12.5 \Rightarrow a = 12.5$

At $t = 10 \text{ min}$,

$\{(a - x) \times 2 + x \times 5\} \propto 32.5$

$2a + 3x = 32.5$

$3x = 7.5 \Rightarrow x = 2.5$

$$K = \frac{1}{t} \ln \left(\frac{a}{a-x} \right)$$

$$K = \frac{1}{10} \ln \frac{12.5}{12.5 - 2.5} = \frac{1}{10} \ln \frac{10}{8}$$

$$= \frac{1}{10} [\ln 10 - \ln 8]$$

$$= \frac{1}{10} [2.303 - 3 \times 0.7]$$

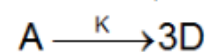
Mo. of g. eq. of R.A.

$K = \frac{1}{10} \times 0.2$

$= 0.02 \text{ min}^{-1}$

A.

For the first-order parallel reactions, starting with 1 M of A, which of the following statement(s) is/are correct?



(A) $[B]_t : [C]_t : [D]_t$ is independent of time.

(B) $[C]_t > [B]_t = [D]_t$

(C) $[C]_\infty = 0.67 \text{ M}$

(D) If $K = 0.5 \text{ min}^{-1}$, then $[B] = 1 \text{ M}$ after 1.386 min.

Ans: A, B, C

$t = \infty$, (when reactant is completely consumed)

$$1 = \frac{p}{1} + \frac{q}{2} + \frac{r}{3} \quad (\text{As per stoichiometry})$$

$$3a + \frac{1}{2} \times 4a + \frac{1}{3} \times 3a = 1$$

$$3a + 2a + a = 1$$

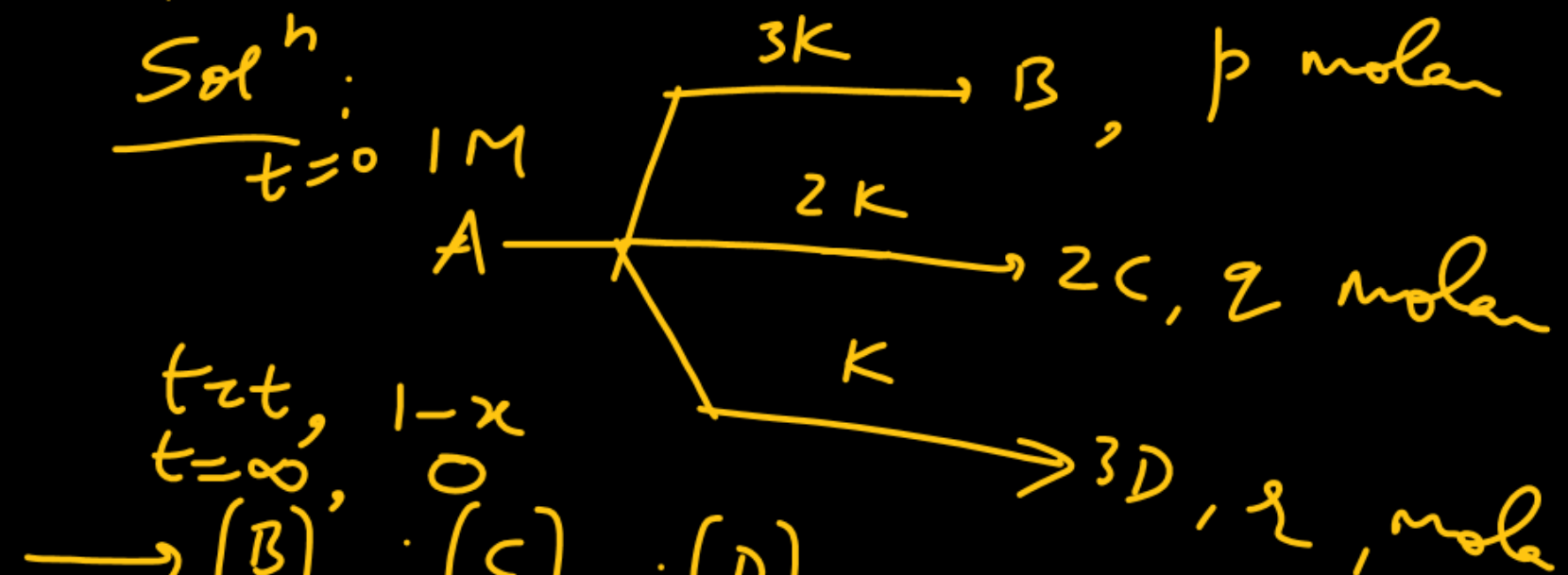
$$a = \frac{1}{6}$$

$$p = \frac{3}{6} = 0.5 \text{ M}$$

$$q = \frac{4}{6} = 0.67 \text{ M}$$

$$r = \frac{3}{6} = 0.5 \text{ M}$$

$$\begin{aligned} 1 \text{ mol } B &= 1 \\ \therefore p \text{ mol } B &= \frac{1}{1} \times p \\ 2 \text{ mol } C &= 1 \text{ mol } A \\ \therefore q \text{ mol } C &= \frac{1}{2} \times q \end{aligned}$$



$$[B]_t : [C]_t : [D]_t = 1 \times 3K : 2 \times 2K : 3 \times K$$

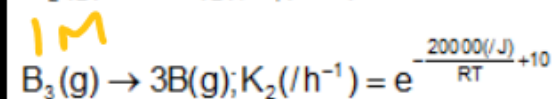
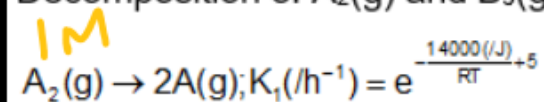
$$= 3 : 4 : 3 \quad (\text{constant} = p : q : r \text{ ratio})$$

(independent of time) $\Rightarrow p = 3a$
 $q = 4a$
 $r = 3a$

$$[C]_t > [B]_t = [D]_t$$

$$\begin{aligned} (B)_{1.386} &= 3K(A)_t \\ &= 3K \times 1 \times e^{-(3K+2K+K) \times 1.386} \\ &= 1.5 \times e^{-3 \times 1.386} \\ &\neq 1 \text{ M} \end{aligned}$$

Decomposition of $A_2(g)$ and $B_3(g)$ follows first-order kinetics as follows.



Here, K_1 and K_2 are the rate constants with respect to disappearance of A_2 and B_3 , respectively. One mole, each of $A_2(g)$ and $B_3(g)$, is taken in a 100 L evacuated flask and at some temperature at which they start decomposing at the same rate. The incorrect information regarding the reactions is

- X (A) The temperature at which the reactions are performed is $\frac{1200}{8.314} K$.
- X (B) At any instant, $\left(\frac{P_{A_2}}{P_{B_3}}\right)$ will be constant and equal to 1.0, where P_{A_2} and P_{B_3} are the partial pressures of gases at that time.
- ✓ (C) At some instant, the total pressure of all gases in the flask may be less than 0.2 atm.
- ✓ (D) At any instant, $\left(\frac{P_A}{P_B}\right)$ will be constant and equal to 1.0, where P_A and P_B are the partial pressures of gases at that time.

$V = 100 L, K_1 = K_2$

$$e^{-\frac{14000}{RT} + 5} = e^{-\frac{20000}{RT} + 10}$$

$$-\frac{14000}{RT} + 5 = -\frac{20000}{RT} + 10$$

$$\frac{6000}{RT} = 5 \Rightarrow T = \frac{1200}{8.314} K$$

$$(Rate)_{A_2} = K_1 \times (A_2)_t = K_1 \times 1 \times e^{-K_1 \times t}$$

$$(Rate)_{B_3} = K_2 (B_3)_t = K_2 \times 1 \times e^{-K_2 \times t}$$

$$K_1 = K_2$$

$$(Rate)_{A_2, t} = (Rate)_{B_3, t}$$

$$(P_{A_2})_t = (P_{B_3})_t \quad \checkmark$$

$$P_i \times 100 = 2 \times 0.0821 \times \frac{1200}{8.314}$$

$$P_i = \frac{0.1642}{8.314} \cong 0.2 \text{ atm}$$

$$(P_t)_{\text{total}} > P_i > 0.2 \text{ atm}$$

~~$\frac{P_{A_2}}{P_{B_3}}$~~

$$\frac{P_A}{P_B} = \frac{2}{3} \neq 1$$

In a study of effect of temperature on reaction rate, the value of $\frac{1}{K} \cdot \frac{dK}{dT}$ is found to be $\frac{1.25 \times 10^6}{T^3} K^{-1}$. Identify the correct statement(s).

(A) The activation energy for the reaction at 250 K is 10 kcal/mol.

(B) The activation energy for the reaction at 2000 K is 1.25 kcal/mol.

(C) The rate of increase of rate constant with the increase in temperature is higher at lower temperature than at higher temperature.

(D) The value of $\frac{d(\ln K)}{dT}$ is 0.625 at 1000 K.

$$\left(\frac{d(\ln K)}{dT} \right) = \frac{2.5 \times 10^6}{T^3}$$

$$= \frac{2.5 \times 10^6}{1000^3}$$

$$= 2.5 \times 10^{-3}$$

$$\neq 0.625$$

Imp

Note:

200 $\rightarrow$ 201 K

300 $\rightarrow$ 301 K

Rate constant will increase more sharply at 200 to 201.

Solⁿ:

$$K = A e^{-E_a/RT}$$

$$\frac{dK}{dT} = A e^{-E_a/RT} \times \left(+ \frac{E_a}{RT^2} \right)$$

$$\frac{1}{K} \frac{dK}{dT} = \frac{E_a}{RT^2} = \frac{1.25 \times 10^6}{T^3}$$

$$(E_a)_{250K} = \frac{1.25 \times 10^6}{T^3} \times R \times T^2 = \frac{1.25 \times 10^6}{250} \times \frac{2}{1000} \text{ Kcal/mol}$$

= 10 Kcal/mol.

(Ea)

2000 K

$$= \frac{1.25 \times 10^6}{2000} \times \frac{2}{1000}$$

$$= 1.25 \text{ Kcal/mol}$$

$$= \ln A - \frac{1.25 \times 10^6}{T^2}$$

$$\frac{d(\ln K)}{dT} = -1.25 \times 10^6 \times \left(-\frac{2}{T^3} \right)$$

$$K = A e^{-E_a/RT}$$

$$\ln K = \ln A - E_a/RT$$

$$= \ln A - \frac{1.25 \times 10^6}{T \times RT}$$

Consider two first-order reactions I and II. The frequency factor of I is 100 times that of II, the activation energy of I is 4.606 kcal higher than that of II. If the ratio of rate constants for the reactions I and II is $x : 1$ at 227°C , then the value of x is $x = 1$

Solⁿ :

$$\frac{K_I}{K_{II}} = \frac{A_I e^{-E_{a1}/RT}}{A_{II} e^{-E_{a2}/RT}}$$

$$\frac{x}{1} = 100 e^{(E_{a2} - E_{a1})/RT}$$

$$E_{a1} = 4.606 + E_{a2}$$

$$(E_{a2} - E_{a1}) = -4.606$$

$$\frac{A_I}{A_{II}} = 100$$

$$R = 2 \text{ Cal mol}^{-1} \text{ K}^{-1}$$

$$T = 227 + 273$$

$$= 500 \text{ K}$$

$$\frac{x}{1} = 100 \times e^{\frac{-4.606 \times 10^3}{2 \times 500}}$$

$$= 100 \times e^{-2 \times 2.303} \Rightarrow x = 1$$

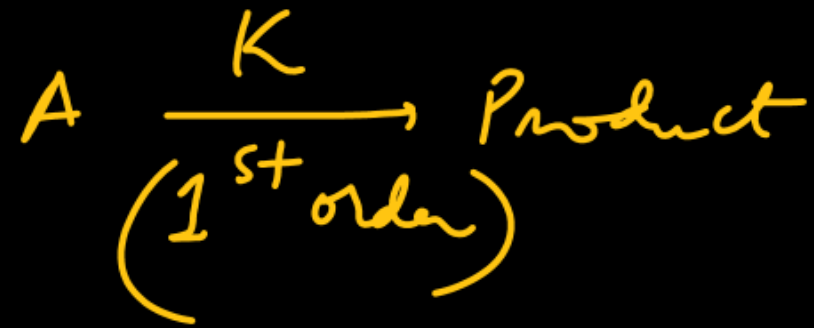
$$= 100 \times e^{-2 \times \ln 10}$$

$$= 100 \times \frac{1}{e^{\ln 100}} = 1$$

JEE-Adv

The rate of a first-order reaction is $0.04 \text{ mol litre}^{-1}\text{s}^{-1}$ at 10 minutes and $0.03 \text{ mol litre}^{-1}\text{s}^{-1}$ at 22 minutes after initiation. The half-life of the reaction (in seconds) is ($\ln 2 = 0.7$, $\ln 3 = 1.1$)

Solⁿ:



$$(\text{Rate})_t = K \cdot [A]_t$$

$$0.04 = \cancel{K} \times \cancel{[A]_0} e^{-10 \times K}$$

$$0.03 = \cancel{K} \times \cancel{[A]_0} e^{-22 \times K}$$

$$\frac{4}{3} = e^{12K}$$

$$\Rightarrow 12K = \ln 4 - \ln 3$$
$$= 1.4 - 1.1 = 0.3$$

$$K = \frac{0.3}{120} = \frac{\ln 2}{t_{1/2}}$$

$$\Rightarrow t_{1/2} = 40 \times 0.7 \times 60 \text{ sec}$$
$$= 28 \times 60$$

$$t_{1/2} = 1680 \text{ sec}$$

$$\begin{array}{r} 28 \\ 60 \\ \hline 00 \\ 168 \\ \hline 1680 \end{array}$$

A hydrogenation reaction is carried out at 500 K. If the same reaction is carried out in the presence of a catalyst at the same rate, the temperature required is 400 K. The activation energy of reaction (in kJ/mol), if the catalyst lowers the activation energy by 20 kJ/mol, is

Solⁿ: $(K)_{500K, \text{ without catalyst}} = (K)_{400K, \text{ with catalyst}}$

$$A e^{-E_a/R \times 500} = A e^{-(E_a - 20)/R \times 400}$$

$$\frac{E_a}{R \times 500} = \frac{E_a - 20}{R \times 400}$$

$$4E_a = 5E_a - 100$$

$$E_a = 100 \text{ kJ/mol}$$

$\rightarrow 100 \text{ kJ/mol}$

